

Whole Foods Isn't Built For Budget Lifters

I wanted to make this visualization because I go to the gym very often and take lifting pretty seriously, so budgeting my meals is something I think about a lot. Getting enough protein can get expensive fast, especially if you're also trying to eat higher-quality food, so I wanted to see which store actually gives the best value. Specifically, I looked at how much protein you get per dollar across different levels of food processing, since I don't want to rely on low-quality protein just to save money. The main takeaway is pretty clear: Target and Walmart are almost identical in terms of value, while Whole Foods consistently gives you less protein per dollar, even at higher quality levels. It basically ends up being a premium option that only really makes sense if you can afford to not care about the price as much.

To create this, I started by cleaning the dataset and filtering out unrealistic values, like extreme prices and outliers in protein per dollar. I then created a new metric, protein per dollar, by dividing protein content by price. Finally, I grouped foods into four bins based on their processing score. Next, I calculated the median protein per dollar for each store within each processing category to make the comparisons more stable and get rid of any potential cluttering of data that could be on the graph. I chose a line plot because position is the clearest way to compare differences across categories, and it makes the gap between Whole Foods and the other stores really obvious. I also added annotations and vertical brackets to highlight how large the gap is, especially for minimally processed foods where the difference is the most extreme. Overall, the design makes the difference in value easy to see right away while still preserving the context of food quality.